MD-0 Cover, goal and governing figures



TODAY: original hall photo (not for measuring)



THE GOAL

The middle hall door becomes a built-in bookcase, flush with the wall, crown and casing untouched. The whole case swings out on hidden pivots: press the right edge, it releases, pull. Behind it: the laundry closet with the LG WashTower.

TODAY

A bathroom pocket door (slides right, into this wall). B the door that becomes the bookshelf. C open to the Urbane Bronze bedroom.

GOVERNING FIGURES

Item	Figure	Source	Status
Opening width W (inside jamb)	28 7/16 in	Magicplan door 6ac582a0.df9703ff	VERIFY IN FIELD
Opening height H (floor to head jamb)	80 7/16 in	Magicplan	VERIFY IN FIELD (finished floor)
Jamb depth J (hall face to closet face)	not recorded	field	MEASURE
Closet inside	36 5/16 x 82 5/8 in	Magicplan room Closet	VERIFY IN FIELD
Ceiling height	121 1/2 in	Magicplan	recorded
Hall clear width at this door	41 1/4 in	Tisa clarification (workbook)	VERIFY IN FIELD
Bathroom pocket door	28 in wide, slides right into wall A-B	Magicplan + PD-01 R4	pocket specialist
Washer/dryer	LG WKG101HVA 27 x 30 3/8 x 74 3/8 in; 1 in sides, 4 in rear; closet vents 48 sq in at >= 6 ft + 24 sq in <= 1 ft	LG manual p.13-14	locked (project instructions)
Case overall depth D	8 1/4 in (shelf 7 3/16 in usable)	Murphy template depth	Tisa to confirm; 9 1/16 alt.
Hinge-side gap gL / free-side gap gR	1/4 / 1 in (Aw 1 1/4)	Murphy illustration	PRELIM: template + mockup
Top + bottom hinge stacks Ah	2 3/8 in	Murphy illustration	PRELIM: measure kit (T + Bt + 1/8)
Case Wc x Hc (with PRELIM gaps)	27 3/16 x 78 1/16 in	computed	cut only from released sheet

SHEETS

MD-0 Cover, goal and governing figures
MD-1 Floor plan, hall and laundry closet
MD-2 Hall elevation of the opening
MD-3 Section A-A through the pivot axis
MD-4 Plan detail: pivot, sweep, latch, stops
MD-5 Case construction and cut list
MD-6 Mark sheet
MD-7 Sequence, hardware and open items

LEGEND

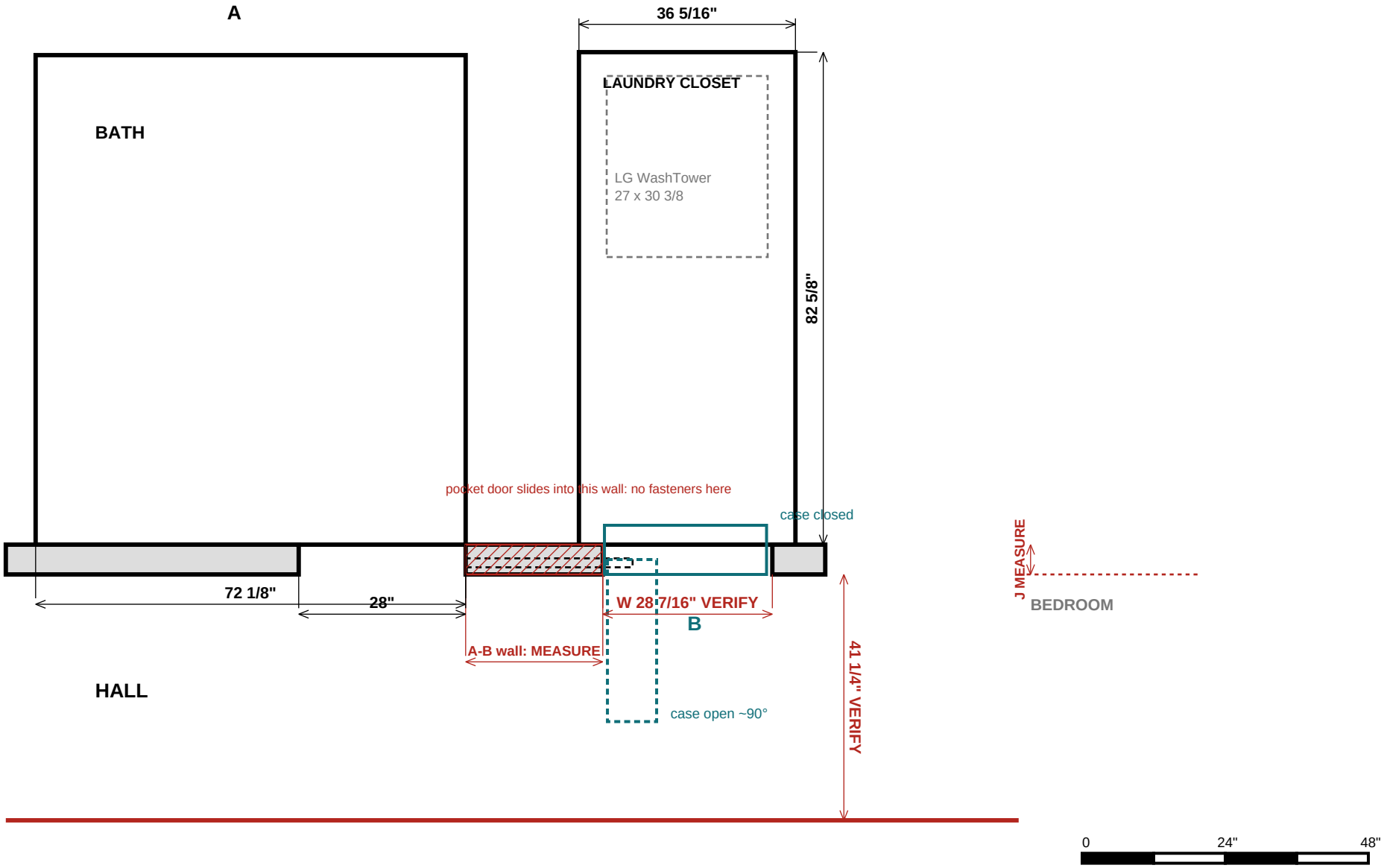
heavy line: existing wall / jamb (cut)
medium: case and parts
light: dimensions
dashed: hidden or open position
teal: new work

red: VERIFY / do not cut
hatch: no fasteners (pocket side)
P: pivot axis
L: push latch
V: vent opening in back

MD-1 Floor plan, hall and laundry closet

reference: hall face of jamb = datum 0

Scale 1:24 (1 in = 24 in). Closet door position within the closet wall and wall A-B length are not in Magicplan: drawn schematic, VERIFY.

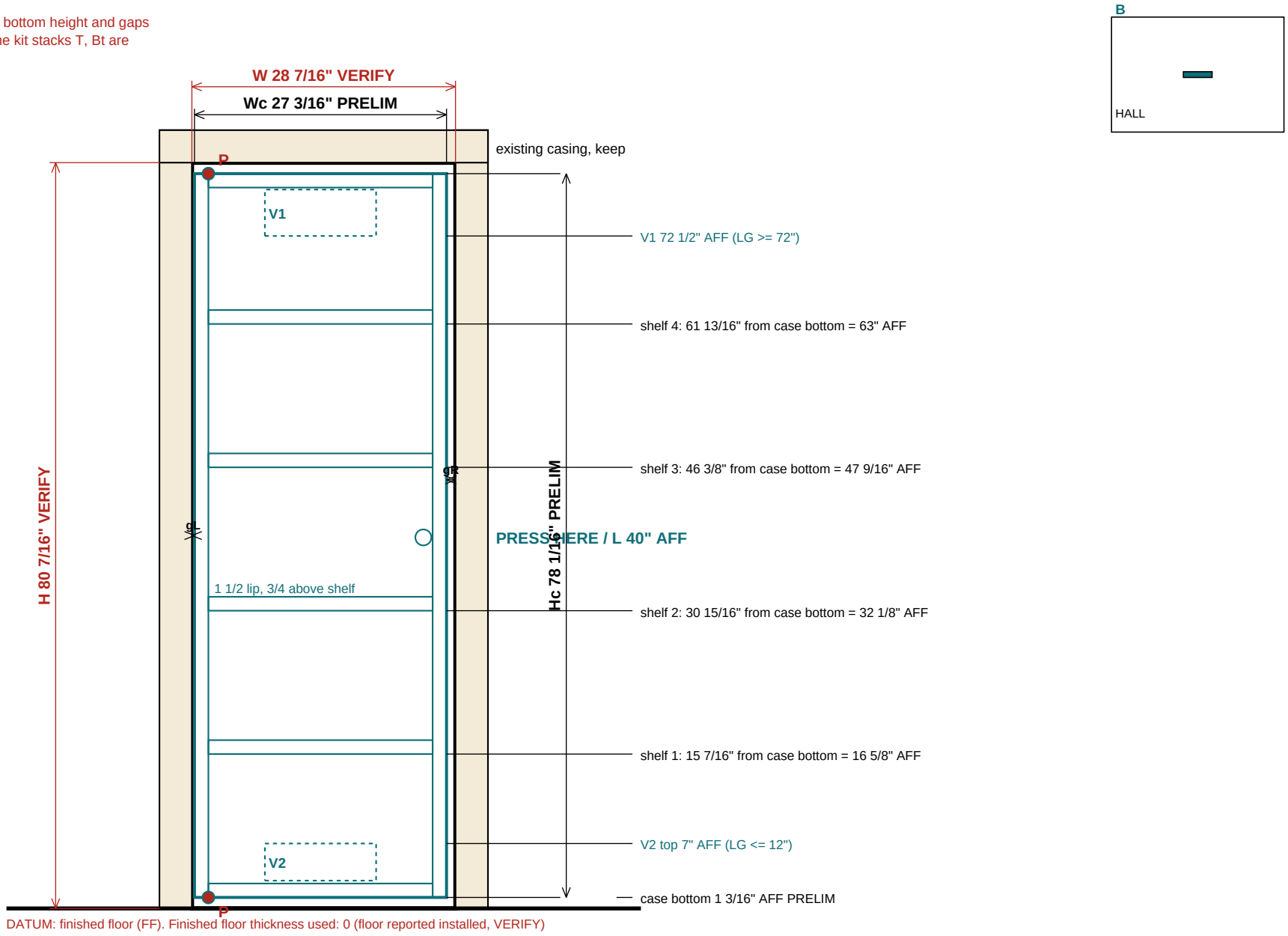


1231 11th Ave, Unit 4 · Murphy bookshelf door (middle hall door B)

Figures: Magicplan plan abb2d2c6 (read 2026-11-02) and LG WKG101HVA manual. VERIFY IN FIELD where marked. English edition governs.
Supersedes MD-02 R3/R4 worked examples (36 x 80 7/8 example retired).

MD-2 Hall elevation of the opening

Scale 1:15.5. Case bottom height and gaps are PRELIM until the kit stacks T, Bt are measured.

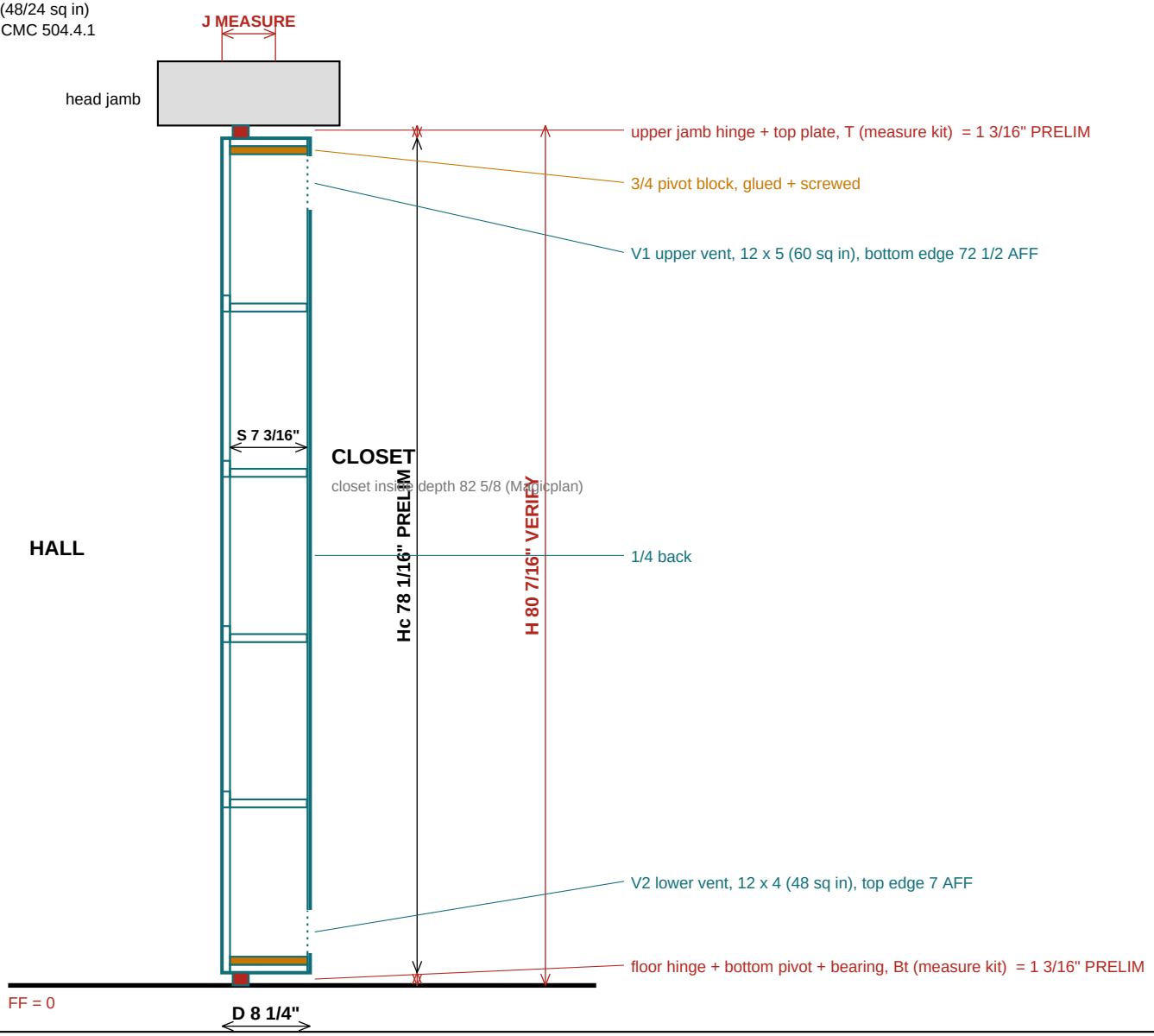


1231 11th Ave, Unit 4 · Murphy bookshelf door (middle hall door B)

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MD-3 Section A-A through the pivot axis

Scale 1:15.5. Back face of case sits D - J inside the closet. LG: 4 in behind the appliance, 1 in at sides; closet depth 82 5/8 gives room. V1/V2 meet LG (48/24 sq in) and total 108 sq in gross for CMC 504.4.1 (100 net).

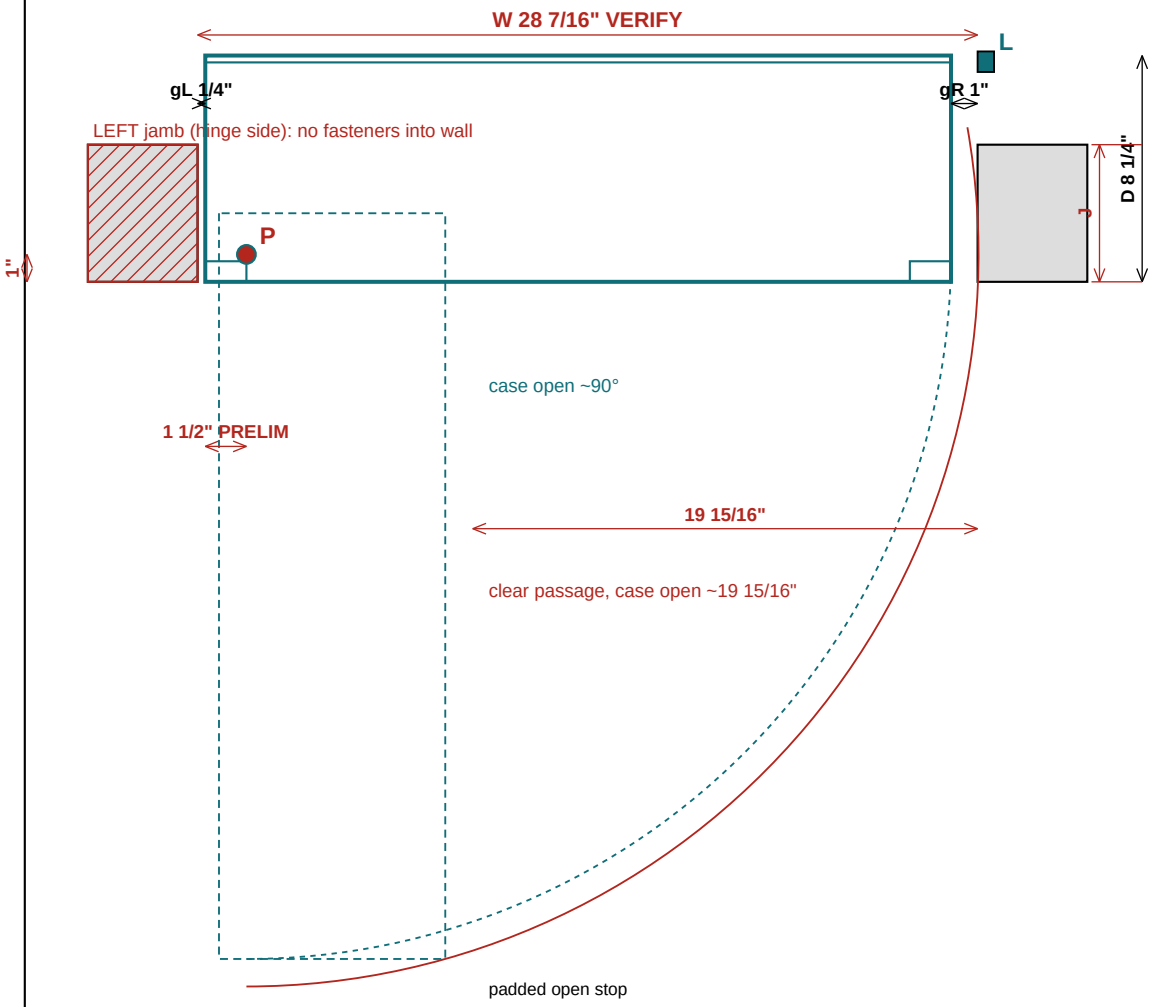


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MD-4 Plan detail: pivot, sweep, latch, stops

Scale 1:7



FREE-SIDE SWEEP CHECK (required gR for the back free corner)

axis from hinge edge	axis from hall face	required gR	required gL
0"	0"	1 1/4"	0"
1"	3/4"	1 1/16"	1/4"
1 1/2"	1"	1"	5/16"
2"	1 1/2"	7/8"	1/2"
1 1/2"	2"	3/4"	1"

Rule: gR >= sqrt((Wc-ax)^2 + (D-ay)^2) - (Wc-ax). At the PRELIM axis (1 1/2, 1) gR must be 1 in: Aw = gL + gR = 1 5/16, Wc = 27 1/8. Template and mockup set the final.

PASSAGE

Case open ~90°, clear between case back and right jamb	about 19 15/16 in	LG unit will NOT pass: remove case
Case removed (pin out, two people)	28 7/16 in clear less floor hinge	LG 27 in wide passes sideways; VERIFY with hinge in place
Case open, projection into hall	about 28 3/16 in of 41 1/4 in hall	hall blocked while open

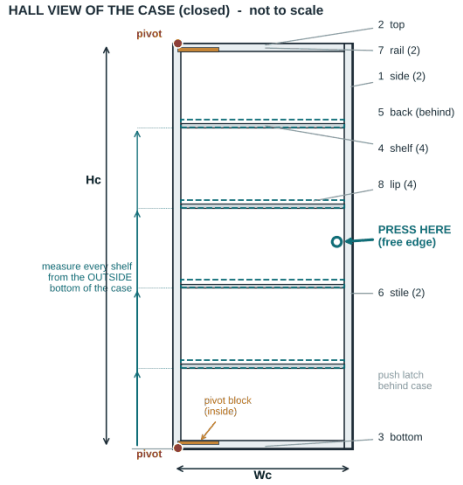
NOTES

P pivot axis, PRELIM 1 1/2 from hinge edge, 1 from hall face of case: SET FROM KIT TEMPLATE + MOCKUP

L ML-ZN80 body on block behind right jamb, strike on case back, 40 AFF; needs 1/8 press travel

closed stop (kit wedge): case still presses in 3/16

MD-5 Case construction and cut list



CASE CONSTRUCTION AND CUT LIST

#	Part	Qty	Thk	Rule	Size with PRELIM gaps (in)	RELEASED
1	Sides	2	3/4	Hc x P	78 1/16 x 7 1/4	
2	Top	1	3/4	(Wc - 1 1/2) x P	25 11/16 x 7 1/4	
3	Bottom	1	3/4	(Wc - 1 1/2) x P	25 11/16 x 7 1/4	
4	Shelves	4	3/4	(Wc - 1 1/2) x S	25 11/16 x 7 3/16	
B	Pivot blocks	2	3/4	8 x P	8 x 7 1/4	
5	Back (with V1, V2 cut)	1	1/4	Hc x Wc	78 1/16 x 27 3/16	
6	Stiles	2	3/4	Hc x 1 1/2	78 1/16 x 1 1/2	
7	Rails	2	3/4	(Wc - 3) x 1 1/2	24 3/16 x 1 1/2	
8	Lips	4	3/4	(Wc - 3) x 1 1/2	24 3/16 x 1 1/2	

Sizes use W, H from Magicplan and PRELIM gaps Aw 1 1/4, Ah 2 3/8. Released sizes come after the kit stacks are measured and the mockup sets the gaps. DO NOT CUT FROM THIS COLUMN.

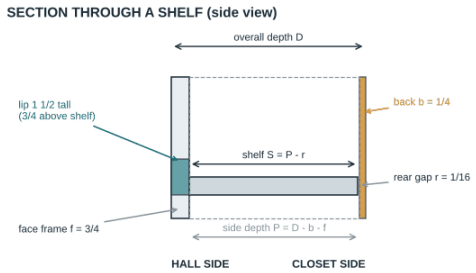
SHELF MARKS from the outside bottom of the case (bottom face of each shelf)

Shelf	1	2	3	4
PRELIM	15 7/16"	30 15/16"	46 3/8"	61 13/16"
RELEASED				

Stock: one 3/4 x 48 x 96 ply (two side strips + three cross strips at 7 1/4), one 1/4 x 48 x 96 back, five poplar 1x2x8. No ripping.

Wc = W - Aw; Hc = H - Ah; P = D - 1; S = P - 1/16; shelf k bottom = 3/4 + k*B + (k-1)*3/4, B = (Hc - 4 1/2)/5 = 14 11/16". Case weight about 53 lb empty; contents limit 150 lb (planning).

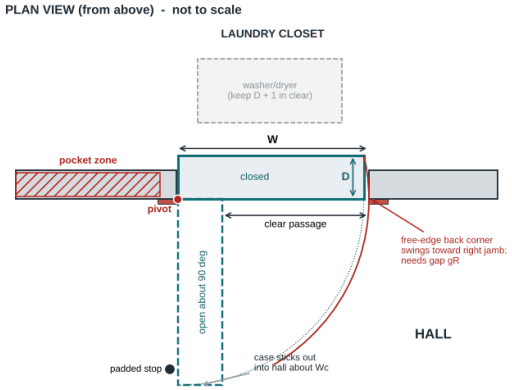
Released by: _____ **Date:** _____ **D:** _____ **Aw:** _____ **Ah:** _____ **Wc:** _____ **Hc:** _____



MD-6 Mark sheet

MARK SHEET: what the crew marks on the jamb and floor

Mark	Where	From what	Distance	Status
M1	Head jamb	inside face of LEFT (hinge-side) jamb	from kit template; PRELIM 1 1/2 + gL	MOCKUP
M2	Head jamb	hall face of jamb	from kit template; PRELIM 1	MOCKUP
M3	Floor	inside face of LEFT jamb	same as M1, plumb within 1/16	MOCKUP
M4	Floor	hall face of jamb	same as M2	MOCKUP
M5	Right jamb, closet side	FF	40 in (latch body center)	RELEASE
M6	Case back	outside bottom of case	V2: 1 3/4 to 5 3/4 up, 12 wide centered	RELEASE
M7	Case back	outside bottom of case	V1: 71 1/4 to 76 1/4 up, 12 wide centered	RELEASE
M8	Case sides (both)	outside bottom of case	15 7/16, 30 15/16, 46 3/8, 61 13/16	RELEASE
M9	Open stop	hall floor or wall	1/2 in before case touches casing or wall, from mockup	MOCKUP



MOCKUP = the cardboard swing test sets this number; RELEASE = the reviewer signs it on MD-5. Mark nothing on the jamb or floor before the reviewer releases MD-6.

Murphy out-swing: upper jamb hinge 4 screws only (skip the two nearest the jamb edge); floor hinge 6; top plate 6; bottom pivot 6, into the pivot blocks. Pin goes in from the front until it sits below the surface.

MD-7 Sequence, hardware and open items

SEQUENCE

1	Hardware check: count kit, find OUT-SWING LEFT templates, measure T and Bt	0.5 h
2	Field check: W, H at three points, diagonals, J, hall width, floor status; type into the Google Doc	1 h
3	Cardboard mockup in the opening: set pivot from template, swing, record gaps, pivot centers, passage	3 h, 2 people
HOLD	Tisa: depth. Reviewer: RELEASED column on MD-5 and MD-6 signed	
4	Cut parts (sides together), label	3 h
5	Assemble square: box, pivot blocks, shelves, face frame, lips; cut V1/V2 in back; fit back	4 h + overnight
6	Fill, sand, prime, two coats Chantilly Lace; mask hinge and strike spots	3 h + drying
7	Prepare opening: remove old door, patch hinge and strike mortises, confirm head jamb solid	2 h
8	Hang: jamb hinges per MD-6, case plates on blocks, stand into bearing, pin from front, 5 cycles	3 h, 2 people
9	Latch and stops: ML-ZN80 at 40 AFF, closed stop with 3/16 travel, padded open stop; 10 presses	2 h
10	Refit trim, caulk to wall only, load test 150 lb, removal practice, photos	2 h, 2 people

HARDWARE

Murphy Door Single Hidden Door Hardware Kit	1	PAID $182.94, order 100070, shipped USPS 10/13
Sugatsune ML-ZN80-BL latch	1	PAID $67.88, Hardware Hut p79677, UPS to 1112 Masonic
3/4 paint-grade ply 4x8	1	check stock
1/4 ply 4x8	1	check stock
Poplar 1x2x8	5	check stock
#8 x 1 5/8 screws, 18ga brads 1 1/4, glue	lot	check stock
Padded open stop	1	needed
Chantilly Lace OC-65 + primer	1 qt ea	sheen open

OPEN ITEMS

Kit delivered to the unit; T and Bt measured	John
Field check W, H, J, diagonals, hall width, floor	John
Pocket-door cavity extent in wall A-B vs this jamb	pocket specialist
Depth 8 1/4 vs 9 1/16	Tisa
Paint-grade ply vs solid oak case	Tisa
Woodwork sheen	Tisa
Vent openings V1/V2 accepted as the closet air path (LG + CMC 504.4.1)	producer / permit

REVISIONS

Rev	Date	
A	2026-11-02	First drawing set from Magicplan figures. Replaces R3/R4 worked examples.